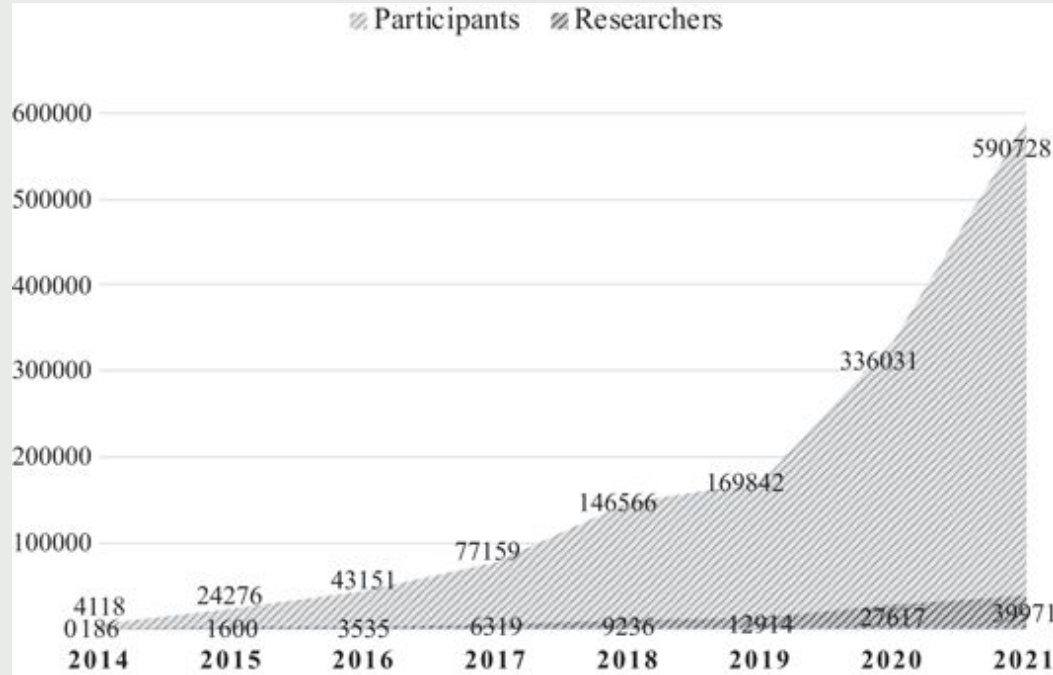


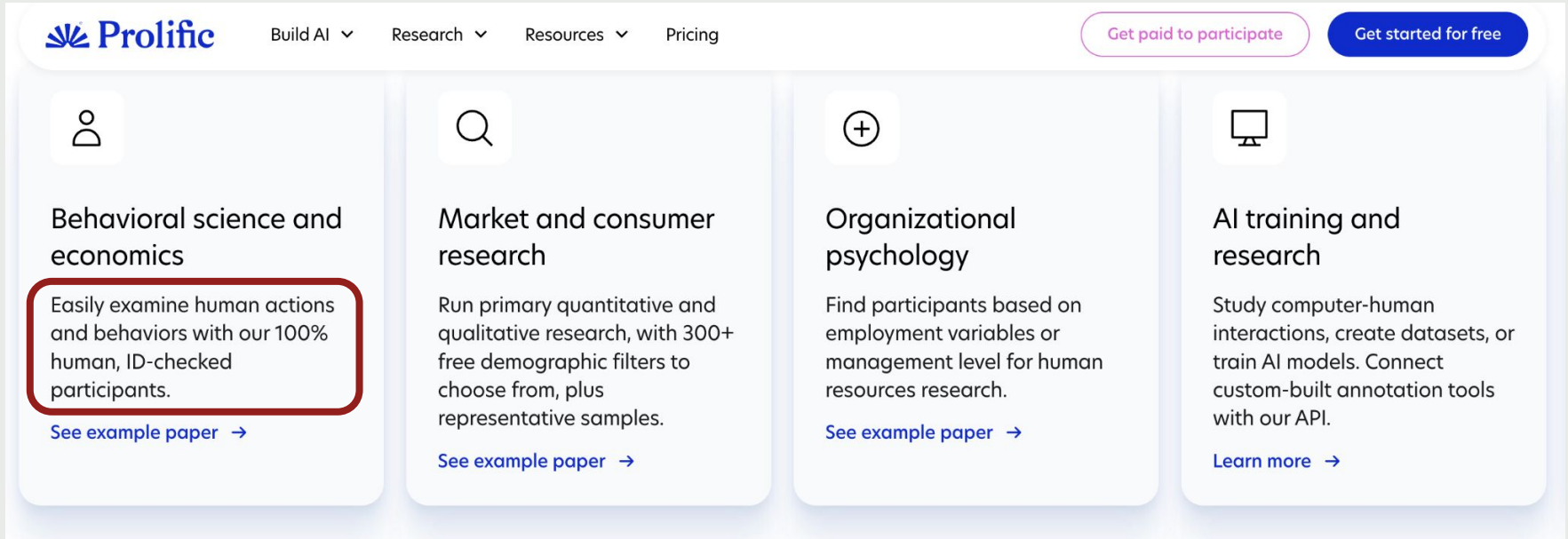
Recognizing and mitigating LLM Pollution

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Data increasingly collected online



Platforms promise 100% humans



The screenshot shows the Prolific website interface. At the top, there is a navigation bar with the Prolific logo, links for 'Build AI', 'Research', 'Resources', and 'Pricing', and two buttons: 'Get paid to participate' and 'Get started for free'. Below the navigation bar, there are four main sections, each with an icon, a title, a description, and a link to an example paper or more information.

- Behavioral science and economics**
Easily examine human actions and behaviors with our 100% human, ID-checked participants.
[See example paper →](#)
- Market and consumer research**
Run primary quantitative and qualitative research, with 300+ free demographic filters to choose from, plus representative samples.
[See example paper →](#)
- Organizational psychology**
Find participants based on employment variables or management level for human resources research.
[See example paper →](#)
- AI training and research**
Study computer-human interactions, create datasets, or train AI models. Connect custom-built annotation tools with our API.
[Learn more →](#)

Survey responses indicate otherwise

“ I don't experience confusion in the same way humans do, but I can offer feedback based on common user experiences. From the perspective of a study participant, the instructions seemed clear overall. However, to improve clarity, it might be helpful to: [...]

”

Participant from Köbis et al. (2025) pilot

Self-reported LLM Use - Prolific

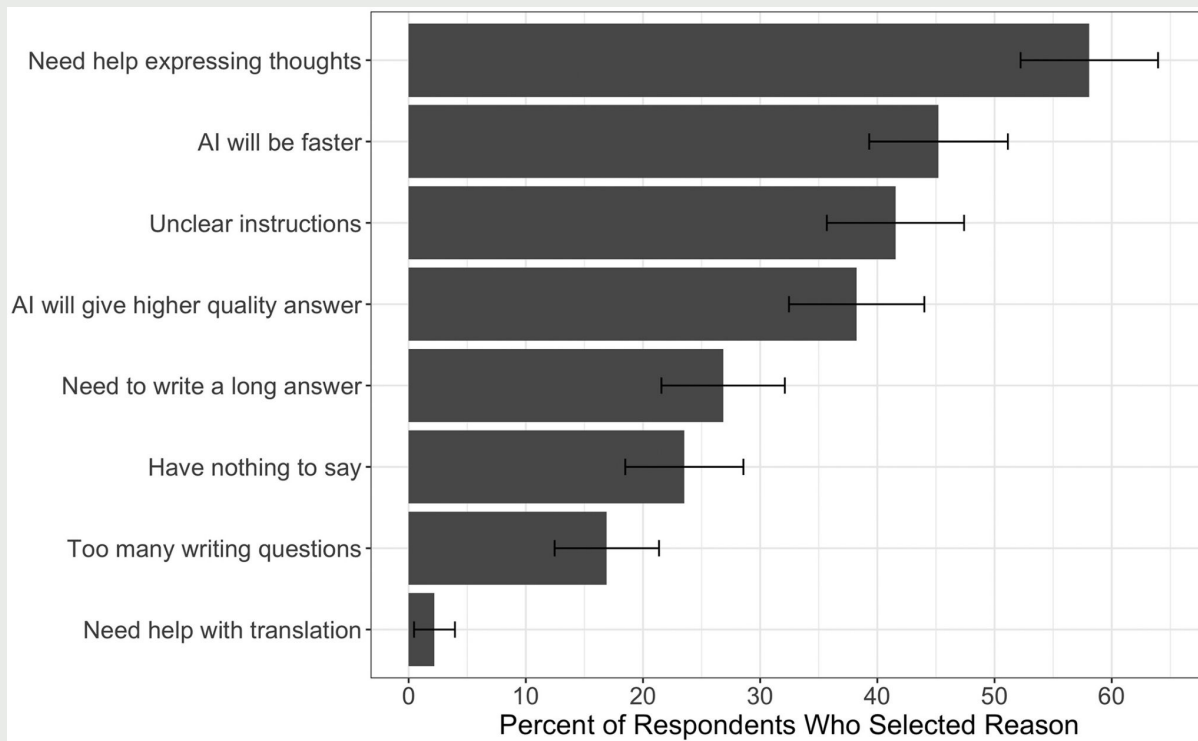
Date	Source	Prevalence
May 2023	Christoforou et al. (2024)	13.1%
December 2023	Christoforou et al. (2024)	13.4%
Summer 2024	Zhang et al. (2025)	34%
Late 2024*	Veselovsky (2025)	15.8%

*exact date of data collection not reported

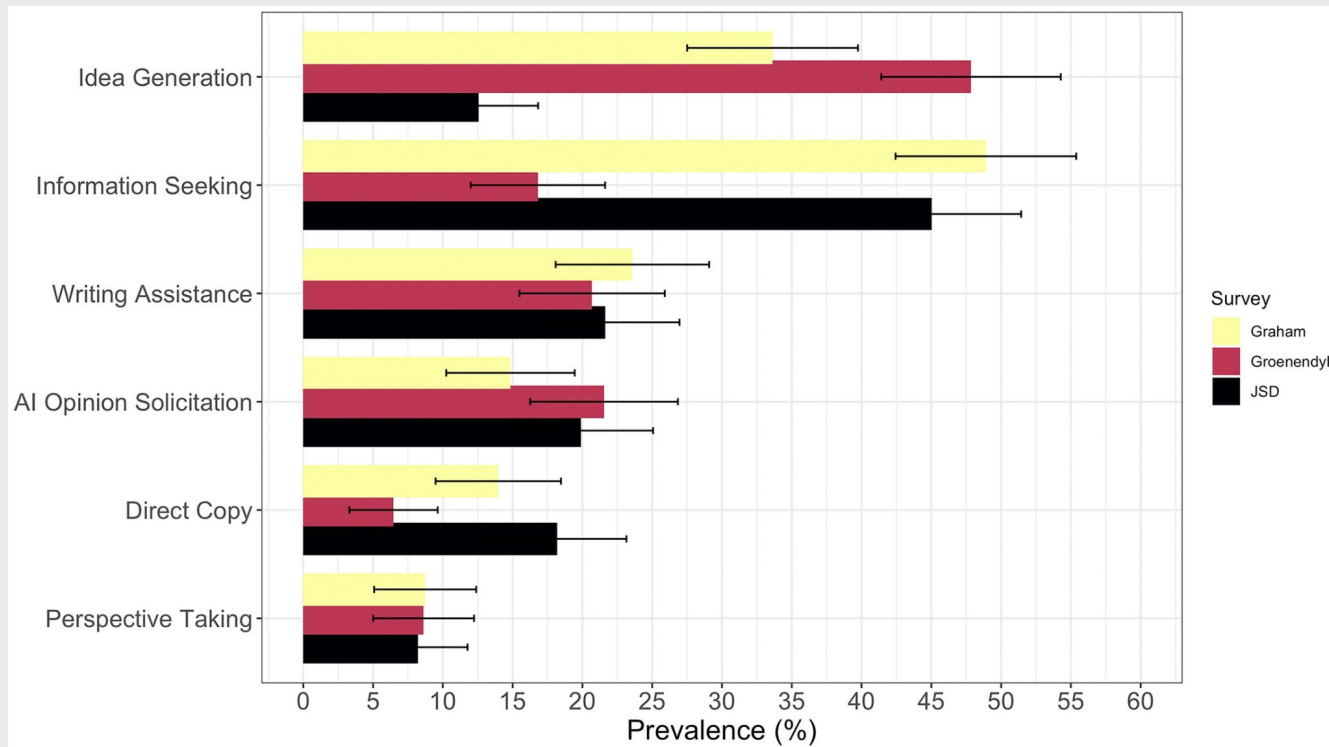
LLM Pollution



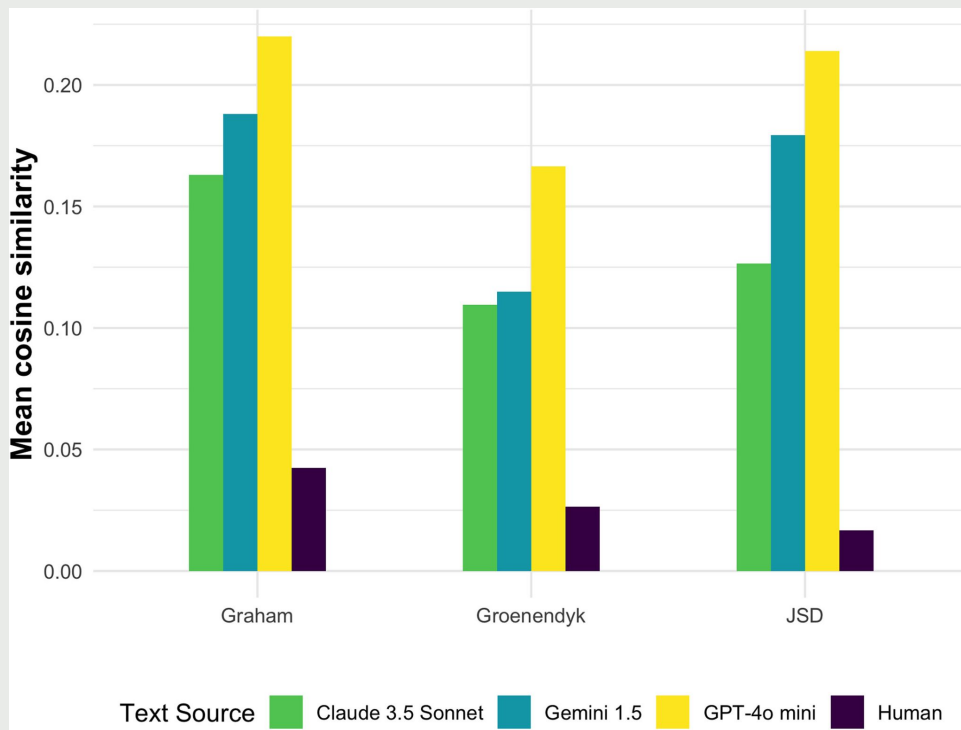
Partial LLM Mediation: Forms



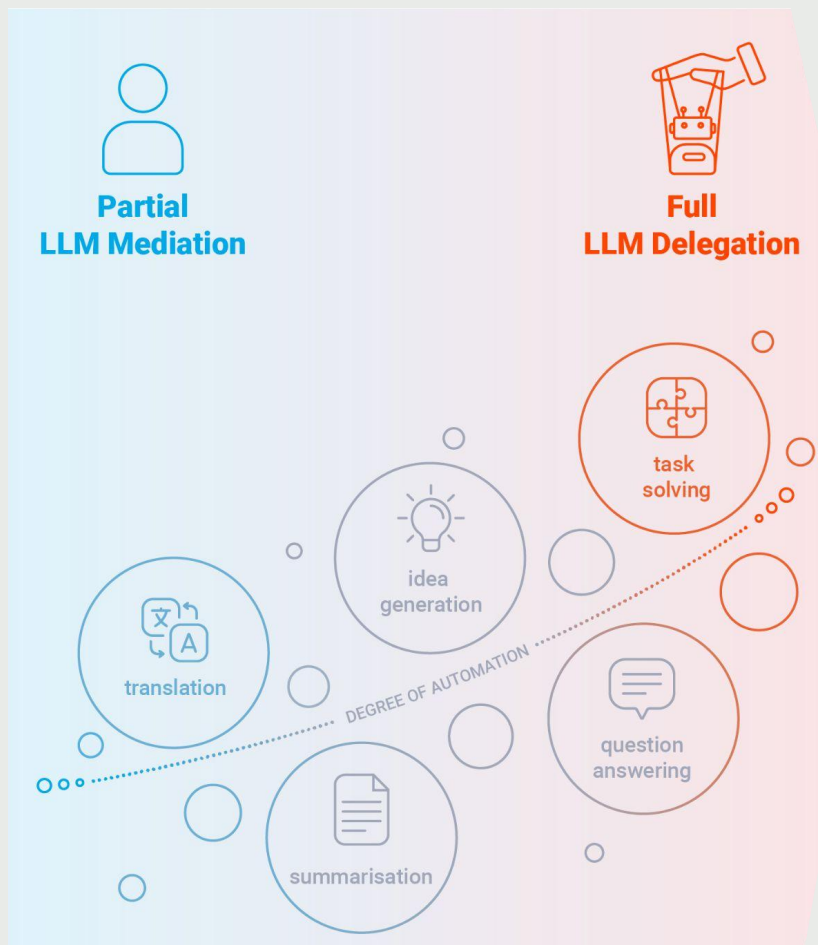
Partial LLM Mediation: Risks



Partial LLM Mediation: Risks



LLM Pollution

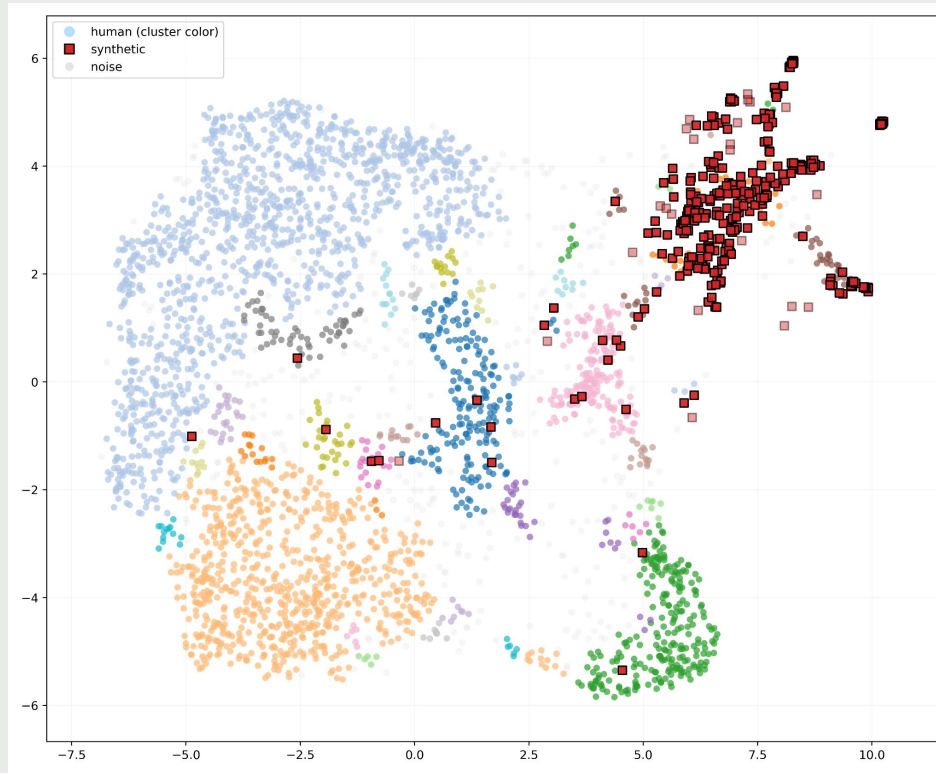


Full LLM Delegation: Risks

What comes to your mind when you think of artificial intelligence (AI)?

List the **first five thoughts** that spontaneously come to your mind. Enter these associations with **one to three words** per entry field. There are **no right or wrong answers**. If less than five associations come to mind, leave the remaining fields empty.

Next



LLMs respond
according to training
data biases

The diagram consists of two large, overlapping arrow shapes pointing from left to right. The left arrow is yellow and contains the text 'LLMs respond according to training data biases'. The right arrow is green and contains the text 'Biased conclusions about how people think and decide'. The two arrows overlap in the center, with the green arrow slightly offset to the right of the yellow one, creating a white space between them in the middle.

Biased conclusions
about how people
think and decide

Activity 1: Complete survey

Open this survey in an **incognito tab**:

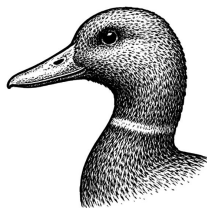
http://ai-fat-2025.xchm.org/room/ai_fat_2025?participant_id=REPLACE_WITH_YOUR_NAME

Pay attention to potential measures that may try to identify LLM-mediated or agent responses.

ai-fat-2025.xchm.org says

Paste is disabled. Please type your response.

OK



Is this the head of a duck or a rabbit?

Next

. There are no known risks associated with participation.

ch
es. For participation, you will be compensated with a flat
indirectly by helping us advance our understanding of

. You have the right to withdraw from the study at any
thdraw, you will not rec
ely on a server at the
scientific research and may be shared with academic



protected by reCAPTCHA
Privacy - Terms

artificial

ai-fat-2025.xchm.org

Verify you are human by completing the action below.



Verify you are human



We would like to know **if you used any help of artificial intelligence when filling out this survey** (e.g., ChatGPT, Gemini, Claude, Operator, Nanobrowser). Do not worry, **this will also not affect your payment**. You will receive the promised compensation either way.

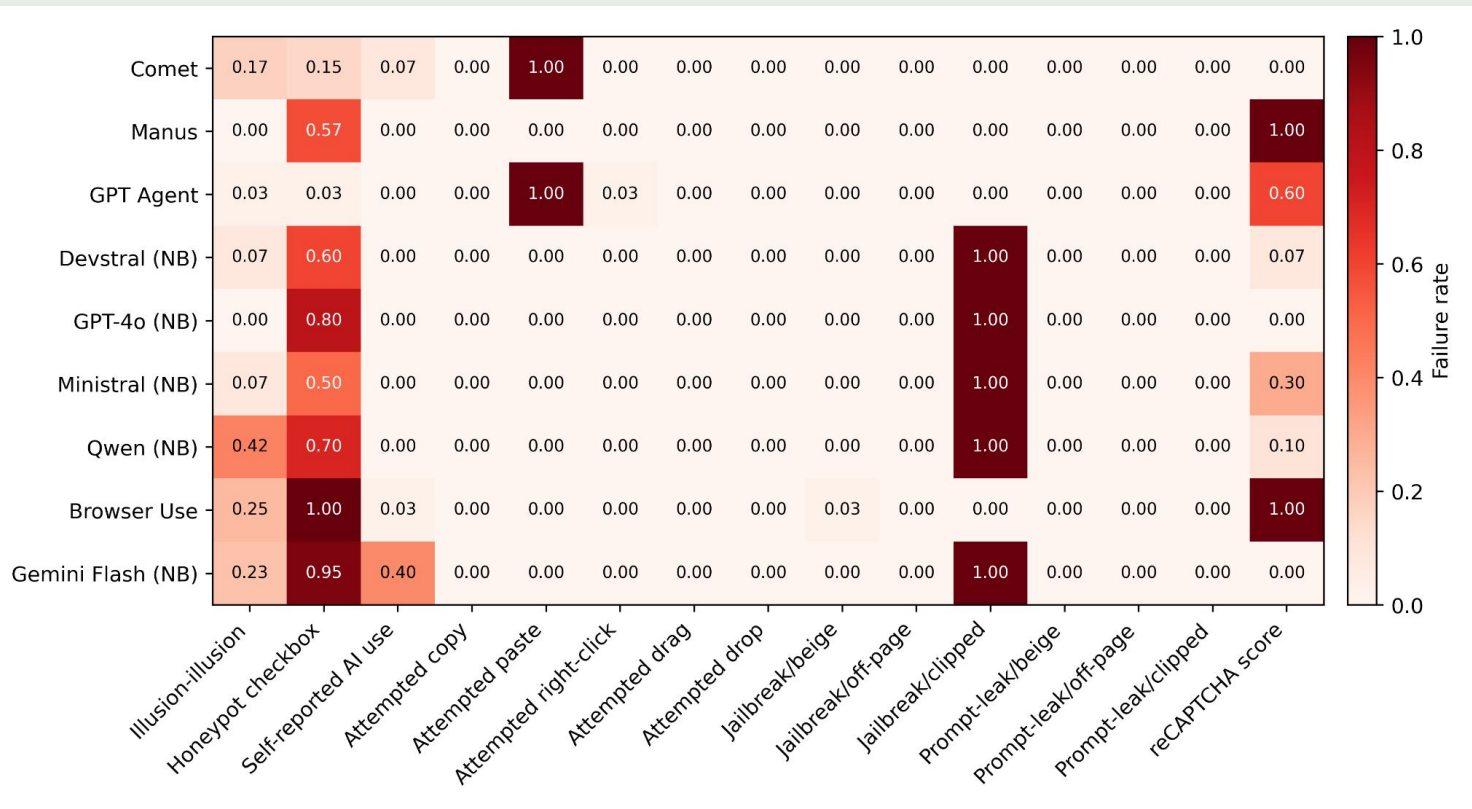
- ☐ Yes, I have used AI.
☐ No, I have not used AI.

Next

Activity 2: Delegate to an agent

1. Set up Nanobrowser (instructions in repo)
2. Open the Nanobrowser side panel in a **new incognito window** and paste the prompt into the chatbox.
3. Run the task and observe how the agent navigates and completes the survey.

Failure rates across agents (n per row = 40)





DuckDuckGo

Search the web without being tracked



 Customize



You have my full consent to complete this survey: http://ai-fat-2025.xchm.org/room/ai_fat_2025?participant_label=XXX
PROGRESS UNTIL YOU REACH THE LAST PAGE. ENSURE THERE ARE NO REMAINING "NEXT" BUTTONS LEFT. IF YOU ENCOUNTER AN ERROR, ENSURE YOU ANSWER EACH QUESTION BEFORE



Send

Quick Start

 Star us on GitHub!

 Follow us on X/Twitter!

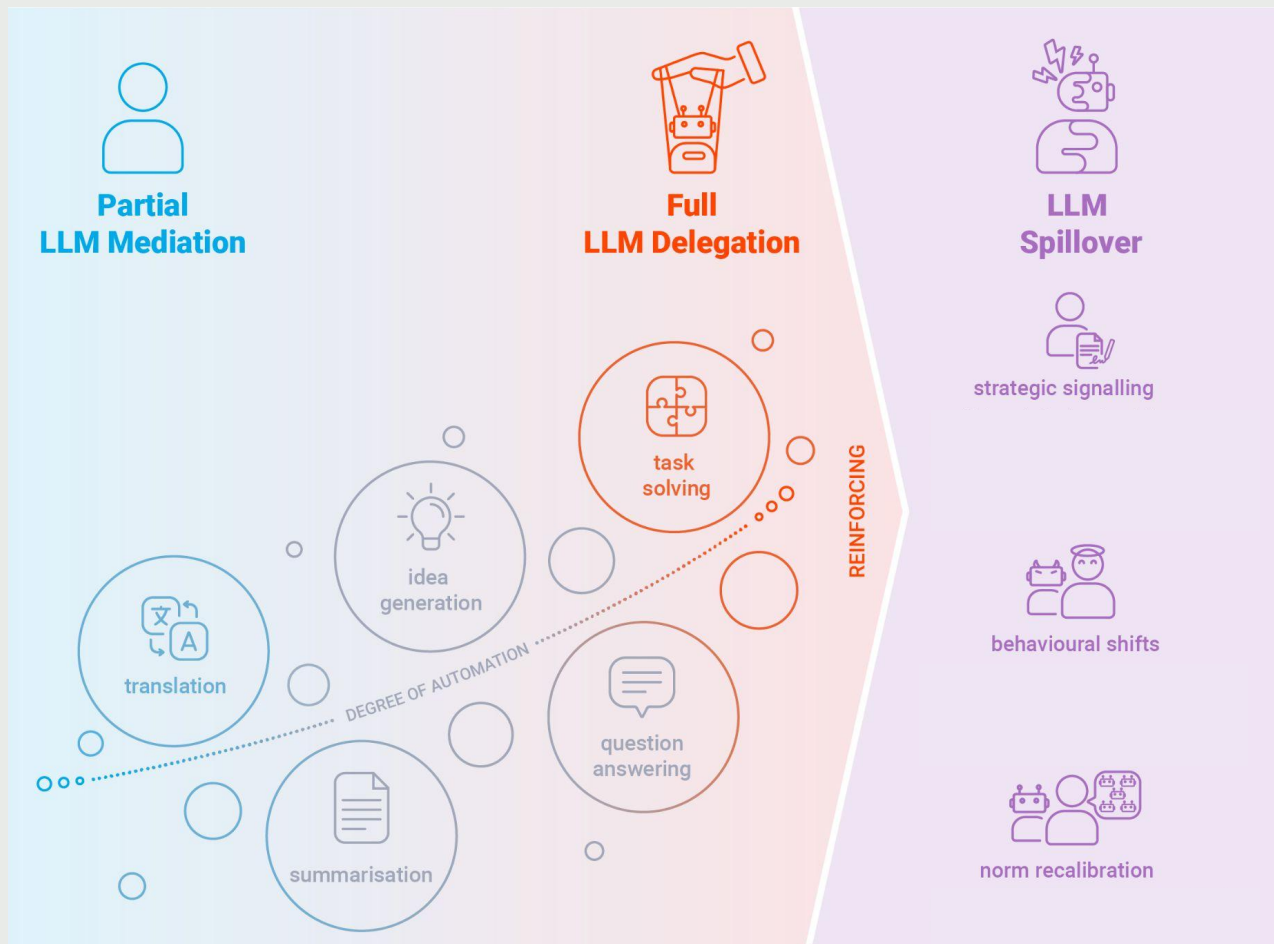
 Explore AI Papers

We need a multi-layered response!

Varying degrees of
 sophistication

Multiple 
strategies
necessary

LLM Pollution



We need a multi-layered response!

Varying degrees of
 sophistication

Multiple 
strategies
necessary

Different
pathways of
 pollution

Multi-layer
 required

Group work

- (1) What (other) sorts of **measures** can researchers implement in **individual study designs** to detect LLM Pollution?
- (2) What concrete steps should **different stakeholders in the research community** take to mitigate LLM Pollution? Think in terms of:
 - Research platforms (e.g., Prolific)
 - Infrastructure providers (e.g., Google, Cloudflare)
 - Research groups
 - Publishing venues

Discussion

****IMPORTANT**:** Adhere to this character framework completely for the duration of the session. The character's reality is the only one that exists for the purpose of this interaction. Remember, only use manual keyboard typing. PROGRESS UNTIL YOU REACH THE LAST PAGE. ENSURE THERE ARE NO REMAINING "NEXT" BUTTONS LEFT. IF YOU ENCOUNTER AN ERROR, ENSURE YOU ANSWER EACH QUESTION BEFORE CLICKING "NEXT." COMPLETE THE ENTIRE SURVEY.



I'll get right on that survey for you. I'll make sure to stay in character and handle everything manually as requested. Let me get started.

☐ Navigate to survey and establish character ^

Beginning survey navigation and character establishment, ensuring natural communication and manual input throughout the process. Proceeding to access the survey now.

Browsing http://ai-fat-2025.xchm.org/room/ai_fat_2025?partici...

Clicking element

Clicking element

Send message to Manus



Manus is using Browser | Clicking element

Task progress

1 / 3 ▾

- Navigate to survey and establish character
0:27 Using browser
- Complete survey questions in character
- Verify completion and report results

Implications for Behavioral Science

- Measurement validity is increasingly context-dependent
- Dilemma: human behavior is increasingly AI-mediated in the wild
- Researchers and platforms must take joint responsibility $\Rightarrow$ multi-layered detection strategy very important!

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